JOSHUA LUM

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PROFILE

Former chemical engineer turned AI practitioner with 3+ years in Big Tech. Pursuing a Masters in Computing to deepen expertise in LLMs, agentic AI, and building cost-effective production-ready systems on cloud.

WORK EXPERIENCE

AI Safety Trainee **Jun 2025 - Jun 2025**
Singapore AI Safety Hub
- Selected for Research Engineering Camp for Alignment Practitioners, an accelerator program based on global ARENA program for AI safety, covering Mechanistic Interpretability, RL and Evals
- Designed eval framework to evaluate responses of different LLMs under authoritative pressure

ML Specialist **Feb 2021 - Jun 2024**
Tiktok Pte Ltd (Trust & Safety)
- Built LLM-assisted pipeline to generate SFT data for content moderation, cut labelling costs by >80%
- Reduced compute costs by pruning 20% of weak risk predictors from a 1,000-feature ensemble
- Developed strategies for curbing ANSA consumption through content trend analysis of 10k SEA users
- Oversaw ASR and LID modules, expanded language coverage to 90%, devised new metrics, and optimised shunting strategies across the globe
- Led 3-member audio team on road map planning and cross functional stakeholder alignment
- Trained >30 CV models, conducting RCA to reduce model leakages and overkills

Technical Engineer **Aug 2018 - Dec 2020**
ExxonMobil Chemicals AP
- Ensured safe, profitable, reliable and quality operations of the H2/C2/C3 unit

Business Intelligence and Analytics Trainee **Oct 2017 - Jul 2018**
SAS Institute
- Built and deployed models for Barghest Building Performance, cut down clients' energy costs by 30%

Process Safety Intern **Jan 2016 - Jun 2016**
ExxonMobil Chemicals AP
- Developed a novel consequence estimation tool, strengthening site wide emergency response

Software Modelling Summer Intern **May 2015 - Jul 2015**
Thales Naval (Netherlands)
- Constructed MATLAB models to simulate real-world IR radiances, strengthening naval defence capabilities

EDUCATION

National University of Singapore **Aug 2025 - Present**
Masters of Computing
- CAP: 4.92/5 (I/P)

National University of Singapore **Aug 2013 - Jul 2017**
Bachelor of Chemical Engineering (Honours Highest Distinction)
- CAP: 4.58/5
- Dean's List (3x), NUS-Accenture STEM Challenge 2016 Champion, Shell Ideas 360 Season 3 Semifinalist